



School: Campus:
Academic Year: Subject Name: Subject Code:
Semester: Program: Branch: Specialization:
Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment :

* Coding Phase: Pseudo Code / Flow Chart / Algorithm

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Proof of Work (PoW)

1. Proof of Work is a consensus mechanism used in blockchain networks to validate transactions.
2. It requires participants, known as miners, to solve complex mathematical puzzles.
3. The first miner to solve the puzzle gets the right to add a new block to the blockchain.
4. This process ensures network security by making it costly and difficult to manipulate data.
5. Miners are rewarded with newly generated coins or transaction fees for their work.
6. PoW consumes large amounts of energy and computing power, leading to high operational costs.
7. Bitcoin and several early cryptocurrencies use the Proof of Work mechanism.
8. Although highly secure, PoW faces criticism for energy inefficiency and environmental impact.
9. It also risks centralization due to dominance of large mining farms.
10. Thus, newer blockchains are exploring alternatives like Proof of Stake for better efficiency.

Proof of Stake (PoS)

1. Proof of Stake is an energy-efficient alternative to Proof of Work.
2. In PoS, validators are chosen to create new blocks based on the number of coins they hold and “stake.”
3. The more coins a user stakes, the higher their chances of being selected to validate transactions.
4. PoS eliminates the need for expensive mining hardware and reduces

power consumption.

5. Validators earn rewards through transaction fees or new tokens for honest participation.
6. If a validator acts maliciously, their staked coins can be partially or fully lost as a penalty.
7. PoS enhances decentralization by allowing more users to participate in consensus.
8. It provides faster transaction processing and better scalability compared to PoW.
9. Ethereum has transitioned from PoW to PoS to improve sustainability and performance.
10. Overall, PoS offers a secure, eco-friendly, and scalable consensus mechanism for modern blockchains.

*** Softwares used**

- Laptop/PC
- Word for documentation
- Brave browser for research

* Implementation Phase: Final Output (no error)

Applied and Action Learning

The implementation involved researching and comparing the working of Proof of Work (PoW) and Proof of Stake (PoS) consensus mechanisms through blockchain whitepapers, technical blogs, and case studies of networks like Bitcoin and Ethereum. The study focused on understanding their operational steps, advantages, disadvantages, and security features.

During implementation, PoW was found to rely on computational effort where miners solve cryptographic puzzles to validate transactions and secure the network. In contrast, PoS selects validators based on the amount of cryptocurrency they stake, reducing energy consumption and hardware dependency.

* Observations

The observations revealed that PoW provides high security and reliability but suffers from high energy usage and slower transaction speeds. PoS, on the other hand, offers greater energy efficiency, faster transaction processing, and improved scalability, though it may face risks of stake centralization.

Overall, the comparative analysis showed that while PoW ensures strong network security through computation, PoS achieves sustainability and efficiency through staking. Both mechanisms play a vital role in maintaining trust and consensus within blockchain systems, each suitable for different network requirements and design goals.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Name :

Regn. No. :

Signature of the Faculty:

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